



February 23, 2026

Thomas Keane, MD, MBA
Department of Health and Human Services
Assistant Secretary for Technology Policy/
Office of the National Coordinator for Health Information Technology
Mary E. Switzer Building
Mail Stop: 7033A
330 C Street SW
Washington, DC 20201

RE: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Dear Dr. Keane,

On behalf of the over 93,000 members of the American College of Surgeons (ACS), we appreciate the opportunity to submit comments to the Department of Health and Human Services' (HHS) Request for Information: Accelerating the Adoption and Use of Artificial Intelligence (AI) as Part of Clinical Care (HHS Health Sector AI RFI) published in the *Federal Register* on December 23, 2025. The ACS is a scientific and educational association of surgeons founded in 1913 to improve the quality of care for the surgical patient by setting high standards for surgical education and practice. The ACS is dedicated to setting the highest standards of care in an optimal and ethical practice environment— and recognizes the transformative potential of advanced technologies in achieving these goals.

The ACS commends HHS for seeking information from the clinical community and other stakeholders to better understand how it can facilitate the acceleration of adoption and use of AI in clinical care. The ACS has been supportive of HHS' efforts to advance the digital ecosystem and AI in clinical care, recognizing that there are clear opportunities to leverage the capabilities of AI for better, more efficient, and affordable healthcare delivery. However, there is also significant risk in implementing AI without clear guardrails and standards for responsible and ethical use, patient safety, transparency, and continuous evaluation and monitoring. Today, structural barriers that delay deployment of validated clinical AI tools also still exist within the digital ecosystem where control sits with electronic health record (EHR) vendors. These barriers can be addressed through promotion of an open, standards-based deployment infrastructure that enables safe, rapid integration of clinically-validated AI tools that empower the clinical community. To fully realize the positive impacts of the integration of AI into clinical care, the Agency must work closely with technical and clinical experts, as well as create a strategic direction that is nimble enough to keep pace with the rapid evolution of digital health technologies. There also must always be a focus on the impact to patient care and outcomes; these

facs.org

CHICAGO HEADQUARTERS
633 N. Saint Clair Street
Chicago, IL 60611-3295
T 312-202-5000
F 312-202-5001
E-mail: postmaster@facs.org

WASHINGTON OFFICE
20 F Street NW, Suite 1000
Washington, DC 20001
T 202-337-2701
F 202-337-4271
E-mail: ahp@facs.org

monumental advancements in healthcare delivery must be for the betterment of patient care and should not detract from the patient experience.

ACS supports HHS's efforts to responsibly accelerate AI adoption in clinical care. While AI holds considerable promise to improve patient outcomes and care delivery, its implementation must be guided by clear guardrails to ensure patient safety, ethical use, transparency, and continuous evaluation. Structural barriers within the digital health ecosystem, particularly those related to EHR integration, continue to delay deployment of validated AI tools. Promoting an open, standards-based infrastructure would enable safe and efficient integration of clinically validated technologies that support clinical decision-making. To fully realize AI's potential, HHS must collaborate closely with clinical and technical experts and establish a strategic framework that remains adaptable to rapid technological advancement. Above all, AI policy must remain grounded in its impact on patient care and outcomes, ensuring that these innovations enhance, rather than detract from, the patient experience.

Procedural AI versus Cognitive AI: A Critical Distinction

As HHS develops regulatory, research, and reimbursement frameworks, the College urges HHS to recognize the fundamental differences between two categories of AI in clinical care: procedural AI and cognitive AI. Procedural AI refers to systems integrated into surgical tools that directly support, guide, or automate elements of operations, often involving robotics or specialized hardware, with the AI component embedded in the device used during the intervention itself. In contrast, cognitive AI consists of software-based systems that analyze data to inform diagnosis and decision-making, providing analytical support but not directly controlling or operating surgical devices during the procedure.

From a surgical perspective, this distinction is critical for policy development. When AI is embedded into tools used intraoperatively, surgeons face unique challenges that distinguish procedural AI from other AI applications. When using cognitive AI outside of the operating room (OR), clinicians can pause to consider additional data, consult colleagues, or seek second opinions when uncertain about or in disagreement with AI recommendations. However, surgeons using procedural AI cannot scrub out of surgery to deliberate when they disagree with AI. They must make real-time—often life-saving decisions—while also considering whether to rely on recommendations from AI-enabled devices during active procedures. This creates a fundamentally different risk profile and decision-making environment. This unique surgical perspective underscores why procedural AI requires regulatory frameworks that account for the time-sensitive, high-stakes nature of intraoperative AI use.

Because of these fundamental differences, a one-size-fits-all approach to AI regulation is inappropriate. For example, procedural AI requires unique post-market evaluation frameworks that account for device malfunction versus user error, which differs significantly from cognitive AI evaluation methods that typically compare outcomes with and without the tool. Procedural AI also requires regulatory efforts focused on transparent device malfunction reporting, clear recall mechanisms, oversight of post-deployment software updates, and appropriate Food and Drug Administration (FDA) labeling that addresses surgery-specific warnings and limitations. Developing effective procedural AI policy necessitates direct collaboration with surgeons and proceduralists who understand these nuances. It is through this lens that the ACS offers our recommendations and comments below.

Key Considerations for Surgery:

The following points represent important topics for the ACS related to regulation; ethics, safety, and transparency; liability; surgeon experts; and reimbursement for AI-enabled tools. We are eager to serve as a resource to HHS as it explores these and other critical areas surrounding AI-enabled tools in surgical practice:

- **Regulation of Procedure-based AI Tools**

The rapid advancement of AI in healthcare has made it difficult for federal agencies to create regulatory frameworks around these technologies. Regulatory strategies should account for the differences across AI categories and use cases, specifically between cognitive applications and AI embedded in procedural technologies. This is critical not only for patient safety but also for the collection of reliable real-world data.

For example, in cases of procedural AI, effective post-market surveillance requires mechanisms that promote transparent reporting of device malfunctions and adverse events, through confidential, anonymized hospital incident reporting systems without penalizing clinicians who use these technologies appropriately. A confidential reporting process is essential to ensuring patient safety, supporting continuous improvement in device design and performance, enhancing the integrity and accuracy of post-market surveillance data, and building trust in these tools and their use in clinical practice. Clear mechanisms for initiating review, recall, or corrective action for AI-enabled tools are also critically important.

Furthermore, FDA labeling for AI/machine learning (ML)-enabled surgical devices warrants careful attention. Required warnings and limitations should clearly describe intended use, known performance constraints, populations in which performance has not been validated, and relevant intraoperative safety considerations. Clear labeling supports informed decision-making by hospitals and surgeons and promotes appropriate clinical use. Maintaining the required warnings and limitations portion of the official FDA-labeling for AI/ML-enabled medical devices is also crucial. Example surgical warnings could include information about the tool misidentifying structures, performance being reduced or untested in certain populations, whether or not the device has been evaluated for robotic surgery, whether the tool ensures achievement of the critical view of safety, and more. Appropriate labeling is critical for vendors, hospitals, and physicians to have discussions about the risks of such tools.

Finally, regulatory frameworks must extend beyond the initial instance of the AI-enabled tools to also address ongoing software updates and adaptive algorithmic modifications that may alter device behavior post-deployment.

Ultimately, the goal of regulatory bodies should be to determine that the AI-based tools are used responsibly and are trustworthy and safe for patient care. Supporting longitudinal tracking of patient outcomes across sites will also be essential for post-market surveillance and AI performance monitoring, which will require mechanisms that give patients verifiable control over their health data as they move across care settings. To achieve this, public-private partnerships that can create a collaborative environment across the technology industry, clinical community, and policy makers are necessary. There must also be a regulatory framework that can quickly transform to keep pace with rapid advancements in the industry. HHS has an opportunity to show strong leadership in this area and would benefit from

creating a standalone entity tasked with convening and coordinating this collaboration with goals of properly governed, trusted knowledge management.

- **Ethical, Safe, and Transparent Use of AI:**

Ensuring ethical, safe, and transparent evaluation and use of AI-enabled tools is foundational to their uptake in surgical care, yet there is a lack of structure around these important criteria.¹ The current environment where transparency requirements are not clearly defined places an enormous burden on surgeons who are expected to rely on AI to minimize errors and advance care yet currently bear responsibility for determining when to defer to or to override AI.² Surgeons need assurance that AI tools are up-to-date, have been rigorously evaluated and validated, and are safe for use within their clinical environment and patient population. Without requirements for transparency around model development, testing, debugging and performance, there is an increased risk of inaccuracy of AI outputs and/or inadvertently perpetuating or exacerbating health disparities. For example, it is important that the system account for a circumstance where an AI-based tool might be used to help a physician diagnose and offer treatment pathways for a patient, but the AI was trained on data taken from a demographic that does not match the patient. Because the AI has never interacted with data that are reflective of the patient, there is significant risk of inappropriate/misdiagnosis and inaccurate treatment recommendations.³ Further, the lack of clear requirements and guidance around transparency and the “black box” nature of the decision-making process of an algorithm makes identifying the potential for these errors extremely difficult for the physician community—especially in instances in surgery where an irreversible, real-time decision must be made intraoperatively. Recently, regulatory bodies have stated that sharing detailed, transparent data around AI algorithms is not useful for clinicians due to the technical nature of the information;⁴ however, we disagree with this assumption. Physicians and healthcare systems must continue to be exposed to and learn to best utilize these data as the uptake of AI-based tools in healthcare becomes inevitable.

We believe that good regulation of these technologies must also consider the level of transparency needed for surgical patients. Transparency to patients includes considerations such as when it is appropriate to disclose when AI is used in their care and how to have conversations with patients. Patients’ perception of AI use in their surgical care varies and can significantly impact their trust in the surgical process. Ensuring patients are comfortable will require effective communication about the contributions of AI, its performance, the privacy and security of their data, and ultimately having the ability to decline the use of AI in their care.³ These examples demonstrate the complex and diverse needs around transparency. Creating a solution that meets the existing and future needs of surgeons and surgical patients will require collaboration and alignment of AI developers, physicians, patients, health systems, policy makers, and more.

The ACS advocates for a trustworthy system, facilitated by the federal government or certified accrediting bodies (or a collaboration of the two), that can ensure AI tools intended for use in clinical care meet technical and clinical care standards of ethical and responsible use, safety, and

¹ Angus DC, Khera R, Lieu T, et al. AI, Health, and Health Care Today and Tomorrow: The JAMA Summit Report on Artificial Intelligence. *JAMA*. 2025;334(18):1650-1664.

² Patil SV, Myers CG, Lu-Myers Y. Calibrating AI Reliance – A Physician’s Superhuman Dilemma. *JAMA Health Forum*. 2025;6(3):e250106.

³ Adegbesan A, Akingbola A, Aremu O, Adewole O, Amamdikwa JC, Shagaya U. From Scalpels to Algorithms: The Risk of Dependence on Artificial Intelligence in Surgery. *J Med Surg Public Health*. 2024; 3:100140.

⁴ 90 FR 60970

transparency, immediately and over long-term implementation. An example of a federal oversight infrastructure for health information technology (IT) is the Assistant Secretary for Technology Policy (ASTP)/Office of the National Coordinator for Health Information Technology (ONC) Health IT Certification Program. To be effective, a transparent certification system should include the following elements, which present areas for opportunity for investment and further development:

- **Technical performance standards for AI applied across its lifecycle** – For AI used in surgical care, such as augmented reality and robotic-assisted surgery, the standards should be specific to the type of tool and could include metrics such as failure to detect proper anatomy, time measurements, technical reliability, accurate registration of images, etc.⁵
 - **Clinical metrics tied to the use of AI** – In surgical care this might include clinical outcomes, such as intraoperative complication rates, patient experience, and patient-reported outcomes. However, this could also include metrics around workflow optimization or detection of intraoperative errors, such as frequency of operating in an anatomical “danger zone.”⁶
 - **Guidelines for the proper human touchpoints** – This enables collaboration between the surgeon and the AI tool across the full episode of care, preserving surgeon autonomy and ensuring surgical judgement is exercised not only at the point of intervention but throughout the patient’s surgical journey, from preoperative planning through postoperative recovery. It is crucial to identify where the surgeon-in-the-loop should be properly applied, which can vary based on use case.
 - **Transparent information for end users** – Requirements must outline what data must be made available to end users to ensure they are informed about the tools and can effectively communicate with patients. An example of this could look like AI Model Cards,⁷ where detailed, transparent information about an AI model’s purpose, design, capabilities/limitations, etc., is shared with end users.
- **The Impact of Regulation on Liability**

Liability related to the use of AI in medicine remains unsettled and continues to evolve. Under existing malpractice law, surgeons are evaluated under the medical standard of care and will typically not be found liable if injury occurs following the standard of care. Surgeons are therefore incentivized to follow this standard no matter what an AI tool might suggest.⁸ As this standard develops over time through the interaction of federal oversight, development of clinical or practice guidelines, payment policy, and patterns of clinical adoption, how HHS approaches AI regulation and program guidance will likely influence how courts assess surgeon conduct in future professional liability actions.

Regulation should also play a critical role in how causation and responsibility are allocated between providers and developers. Many AI systems, particularly those using opaque or “black box” methodologies, do not make their reasoning transparent to the end user. Dataset bias, performance drift,

⁵ Kos TM, Colombo E, Bartels LW, Robe PA, van Doormaal TPC. Evaluation Metrics for Augmented Reality in Neurosurgical Preoperative Planning, Surgical Navigation, and Surgical Treatment Guidance: A Systematic Review. *Oper Neurosurg*. 2024;26(5):491-501.

⁶ Mascagni P, Alapatt D, Urade T, et al. A Computer Vision Platform to Automatically Locate Critical Events in Surgical Videos: Documenting Safety in Laparoscopic Cholecystectomy. *Ann Surg*. 2021;274(1):e93-e95.

⁷ Coalition for Health AI. Applied Model Card. Accessed February 17, 2026. <https://www.chai.org/workgroup/applied-model/model-card-and-registry>

⁸ Price WN, Gerke S, Cohen IG. Liability for Use of Artificial Intelligence in Medicine. In: Solaiman B, Cohen IG, eds. *Research Handbook on Health, AI and the Law*. Edward Elgar Publishing Ltd; 2022.

and model degradation are often not visible to end users, and surgeons generally are not in the best position to independently validate AI outputs in real time. Without clear guardrails or liability protections for the provider, these characteristics risk misalignment in liability allocation, with responsibility falling on clinicians even when adverse outcomes stem from system design, data limitations, or integration decisions. Thoughtful regulatory oversight can mitigate these risks and more appropriately align them with existing frameworks such as those for medical device product liability.

In addition, the integration of AI into surgical care raises important questions regarding informed consent. As AI-enabled tools increasingly influence operative planning or execution, patients may reasonably expect to understand when such technologies are being used and the role they play in their care. However, surgeons cannot be expected to disclose technical details regarding algorithm development, training data, or system architecture that are beyond their knowledge or control. Federal guidance clarifying appropriate disclosure standards for AI-enabled surgical tools would help align informed consent obligations with realistic clinical practice, ensuring patients receive meaningful information without imposing impracticable burdens on individual clinicians.

Distinct Considerations for Procedural AI Liability

Unlike cognitive or clinical decision support (CDS) tools, which provide recommendations to inform professional judgment, procedural AI systems directly influence or participate in the execution of surgical care. When a procedural AI system contributes to tissue injury or affects surgical mechanics, causation may appear directly attributable to the surgeons' actions, specifically, whether intervention was feasible and whether a reasonably prudent surgeon could have acted differently. When a procedural AI system operates within its labeled indications, executes functions as designed, and produces outputs or actions based on embedded algorithms that the surgeon cannot meaningfully test or recalibrate in real time, responsibility should not default to the surgeon solely because an adverse event occurred in the OR.

Concerns around liability are further heightened where procedural AI systems are classified or treated similarly to advisory CDS tools and thereby avoid robust premarket evaluation or ongoing oversight by the FDA. In other words, these non-device tools may be framed in litigation as clinical aids rather than as independently accountable medical devices. In such circumstances, developers and institutions may face limited accountability, while liability risk shifts downstream to surgeons who did not design, train, or update the system and may have limited ability to evaluate its internal functioning. Even in cases where surgeons have algorithmic transparency, this access alone should not automatically invoke liability.

The ACS urges HHS to recognize procedural AI as distinct from advisory CDS tools and to align oversight and liability principles more closely with established medical device regulatory and products liability frameworks.

First, procedural AI warrants stronger upstream oversight. Because these systems can directly cause physical harm and may involve data- or design-driven error modes that surgeons cannot always override in real time, federal review and oversight frameworks should reflect their heightened risk profile. Clear labeling, defined user responsibilities, and meaningful post-market performance monitoring are essential to protect patients and ensure appropriate accountability.

Second, liability should track meaningful control. Surgeons should not bear responsibility for harms arising from procedural AI behaviors that were not reasonably foreseeable or preventable through

ordinary surgical vigilance. Responsibility should align with control, knowledge, and the practical ability to intervene, not merely with the surgeon's presence in the OR.

Third, HHS should consider mechanisms to protect compliant use. Surgeons who use AI systems in accordance with FDA-labeled indications, exercise independent clinical judgment, and document reasonable reliance on, or departure from, AI outputs should not face disproportionate liability exposure. While use of an FDA-authorized product consistent with its labeled indications does not provide absolute immunity from malpractice claims, it may serve as evidence that the provider adhered to established, evidence-based standards of care. Federal guidance supporting compliant use would promote patient safety while encouraging responsible innovation. Clear, coordinated federal policy in this area will help ensure that liability principles support, rather than hinder, the safe and effective integration of AI into surgical practice.

- **Leveraging Surgeon Experts**

The ACS requests that HHS build pathways for collaboration with surgeons and specialty societies; when AI is developed for use in surgical practice, engagement of the surgical community must be the rule, not the exception. To ensure the safety, efficacy, and appropriateness of surgical AI tools, surgeons must be involved from the beginning of the research and development lifecycle to implementation, and through evaluation of ongoing performance of AI. Only surgeons can determine if the AI tool achieves the intended clinical outcome of a procedure—especially when considering procedural AI. Therefore, surgical specialty societies should define the goals of care for a surgical AI tool and ultimately determine if those goals are being met. It is also important to note that in today's environment, the technology, alone, is no longer the main barrier to realizing the potential of surgical AI tools. Instead, current hurdles include issues with implementation, change management, administrative buy-in, and more, making the perspective of practicing clinicians, such as surgeons, essential.

Developing and maintaining partnerships with the surgical community will help to ensure AI-based tools and all supporting data are high quality, safe, and appropriate for clinical use. The ACS has extensive experience leveraging clinical data for improved healthcare through its clinical data registries and can offer validated, high-quality data for research and development activities. With its distinguished membership, broad surgical expertise, and network of specialized committees, the ACS is ideally positioned to partner with HHS in advancing AI-enabled tools for surgical care. The ACS looks forward to future opportunities to partner with the HHS on this important work.

- **Reimbursement**

A primary challenge to appropriate reimbursement is the absence of clear coding pathways for AI-enabled tools used by physicians, as many applications do not align with existing Current Procedural Terminology (CPT) or Health Care Common Procedure Coding System (HCPCS) codes. The ACS urges HHS to support the development of coding pathways that recognize physician-directed use of AI, with reimbursement reflecting surgeon oversight as necessary rather than payment for software use alone, when appropriate.

Revaluation of existing surgical services presents an additional concern, as AI adoption may be assumed to reduce physician work, resulting in inappropriate downward pressure on reimbursement. In practice, AI often shifts or increases physician work, as surgeons must oversee the integration, validation, and safe use

of AI-enabled procedural technologies throughout the surgical episode of care. The ACS therefore recommends that HHS require robust empirical evidence and meaningful clinician input before using AI adoption as justification for revaluing surgical services and consider new models for reimbursing physicians for the increased work associated with the use of new technology. Depending on decisions made on the issue of professional liability, this could also warrant methodological changes to valuing professional liability insurance relative value units (PLI RVUs).

To address delays inherent in legacy payment systems, the ACS further encourages HHS to consider time-limited, evidence-generating coverage pathways for AI tools with strong preliminary evidence of clinical benefit, allowing conditional reimbursement while requiring ongoing data collection prior to permanent coverage and payment decisions.

The ACS appreciates the opportunity to comment on these important issues, and we look forward to continuing dialogue with HHS on issues impacting AI/ML in surgical care. Please contact Vinita Mujumdar, Chief of Regulatory Affairs, at vmujumdar@facs.org, or Jill Sage, Chief of Quality Affairs, at jsage@facs.org, with any questions.

Sincerely,



Patricia L. Turner, MD, MBA, FACS
Executive Director and CEO